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M 302 TURBINE BYPASS SYSTEM

1. SCOPE OF WORK

Complete Turbine Bypass System for each steam pressure level system of the HRSG as required for a dual or triple pressure configuration, **with reheat as applicable.**

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Each system shall be complete with piping, valves, support steel, insulation, instrumentation and controls, and all other appurtenances for a complete and operable system.

This shall include but not be limited for each pressure level to the following:

- Bypass piping system.
- Complete desuperheating system.
- One (1) 100% Steam flow isolation and control valves.
- One (1) desuperheater water flow control valves.
- Support steel.
- Thermal insulation and lagging.
- Miscellaneous piping and valves.
- Safety valves. According to the Part 4 Section 3, M-004 Valves.
- Silencer where required.
- All connecting nozzles and bosses for vent, drain, warm up, sampling, test, instruments, branch line etc. connections.
- In line devices such as flow elements and thermowells.
- Complete instrumentation and control systems and devices.
- Complete electrical equipment as specified in **clause 8 herein.**

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 “Codes and Standards”**. The codes and standards of following organizations extracted from **Part 4, Subsection 2.5 “Codes and Standards”** or international equivalent standards are specifically applicable to the design manufac-

turer and testing of equipment included in this specification.

ANSI B31.1	American National Standards Institute.(latest edition including Addendum)
ASTM	American Society of Testing and Materials.
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HEI	Heat Exchange Institute
ICEA	Insulated Cable Engineers Association.
IEC	International Electrotechnical Commission.
IEEE	Institute of Electronic and Electrical Engineers.
JIS	Japan Industrial Standard
NEMA	National Electrical Manufacturer's Association

3. DESIGN AND OPERATING CONDITIONS

The operating and design conditions shall be as follows:

The turbine bypass systems shall be designed to safely and reliably control the flow and temperature of the steam bypassing the turbine at all pressure levels during steam turbine trip and during major system load rejections so that the HRSG can be safely kept on line. For a major system load rejection, the turbine bypass system shall be capable of coordinating the steam turbine and HRSG such that the steam turbine can carry only house load and the HRSG are kept on line. The size of the system and the speed of operation of the steam bypass system shall prevent the lifting of the HRSG safety valves in case of start-up and of load rejection from any Combined Cycle-load.

In addition to the above, the turbine bypass systems shall be capable of controlling the flow and temperature of the steam bypassing the turbine during HRSG start-up and prior to start-up of the steam turbine set for both cold and hot start conditions. In this mode of operation, the system shall enable the operator to load up the HRSG to a stable condition and bring steam pressures and temperatures to the values required for turbine metal temperature matching purposes.

4. DETAILED REQUIREMENTS

4.1 General

Fully integrated steam bypass systems shall be installed to allow

- Rapid matching of steam and turbine metal temperatures during start-up in order to reduce both the time of the start-up sequence and thermal stresses on the turbine.
- The HRSG to continue generating at reduced load after a load rejection on the steam turbine generator without the HRSG safety valve operating.

The turbine bypass systems shall be designed to take account of all anticipated operating conditions. The bypass system shall be provided to direct steam flow to condenser.

The rating of the systems shall allow the HRSG to be brought to full output while the steam generated is directed through the bypass system to the condenser.

The bypass pressure reducing and desuperheating valves, dump tubes, spray water modulating control equipment and instrumentation, necessary bypass control initiating devices and instrumentation, together with spray and steam bypass piping, shall be furnished by the Bidder. Quick acting isolating block valves shall be provided upstream of the bypass valves to allow protection of the condenser in the event of failure of the bypass control valves.

The pressure reducing valves operating under upstream pressure responsive control and the spray water desuperheater operating under downstream temperature/pressure responsive control shall reduce the conditions of the steam leaving the bypass system to conditions acceptable to the receiving component supplier.

The bypass valves shall be controlled to a set point relating to the respective main steam pressures, modified as appropriate to suit both constant and variable pressure modes of operation.

Interlocks shall prevent the opening of the steam bypass valves unless the following parameters have been established

- Spray water at the correct pressure is available at the pressure reduction desuperheating station and/or at the dump devices.
- Condenser seawater cooling pumps are operational and condenser back pressure is below maximum allowable set point.
- Condensate pumps are in service.

The bypass control and isolating valves shall close rapidly upon detection of low spray water pressure, high condenser back pressure, or high steam temperature downstream of the desuperheating station. For this purpose and for accurate control bypass control and isolating valves shall be provided.

Warming drains shall be provided as close to the bypass system block valves as possible in order to minimize thermal shock on the bypass valves and upstream pipework when the bypass comes into operation.

The sizing, positioning, and acoustic treatment of the pressure reducing and desuperheating valves, pipework and other associated equipment shall be provided to reduce noise levels within this limits of Part 4, Section 2.9.

Access to the bypass valves by stairs or walkways for inspection or maintenance purposes shall be provided.

The Bidder shall be responsible for the complete design of the bypass system and shall size it to cope with the required duty under all operating conditions, giving due consideration to the requirements to meet the turbine metal temperature matching over the full load range and load rejection stability.

The system shall be of proven design backed by field experience over multiple installations of similar size operating over a number of years.

4.2 Control Valves

The technical requirements for control valves are as specified in **Part 4, Section 3, M-005 “Control Valves”**.

4.3 Isolating Valves

The Bidder shall furnish fast acting isolating valves for the turbine bypass system, capable of being closed during operation of the bypass system to avoid damage of the downstream components due to failure of pressure and temperature reduction function..

4.4 Desuperheater

The Bidder shall furnish means for deuperheating the bypass steam which may be by injection of into the steam line or into the body of the steam control valves.

5. CONTROL RANGE

The Bidder shall ensure that the turbine bypass system will be able to bring the turbine from valves wide open plus 5 % overpressure condition to house load without lifting HRSG safety valves or exceeding allowable HRSG water level limits. This shall be accomplished without undue stress or loss of fatigue life to any of the equipment. The Bidder shall perform on-line tests to demonstrate this control range to the satisfaction of the Owner. Remedial action including, but not limited to, increasing the size of the bypass system shall be taken by the Bidder at no additional cost to the Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Bidder shall conform to the quality control requirements of Part 4, Subsection 2.7.

The quality control procedures shall include but not be limited to the following.

- Welding.
- Testing.
- Continuity and wiring tests.
- Insulation and high potential tests.

The quality control documentation shall include but not be limited to the following

- Shop Test Reports.
- Material Certifications.
- Results of all Non Destructive Testing.
- All other documentation required by applicable codes and standards.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply installation and commissioning of complete instrumentation and control systems and devices for the turbine bypass system, including but not limited to those listed and described.

7.2 Scope

The bypass control system shall be implemented in the DCS. The system shall include but not be limited to:

- Control and monitoring necessary for the operation of turbine bypass system.
- System and termination cabinets installed in the electronic equipment room as part of the DCS.
- Local mounted instrument panels.
- All field devices required for a complete system including but not limited to the following:
 - (1) Pressure indicators, transmitters and switches.

- (2) Temperature elements, switches and wells.
- (3) Flow elements and transmitters.

7.3 Functional Requirements

In addition to the requirements listed in above the control system shall monitor the status of all equipment and process parameters associated with the operation of the turbine bypass system and shall provide the necessary permissives for control purposes, and alarm outputs for alarm annunciation system. The critical control parameters shall be applied with redundant system.

The turbine bypass control system shall interface with the HRSG control system, the steam turbine generator digital electro-hydraulic control system, the unit coordinated integrated control system and the unit safety trip system. The Bidder shall furnish the Owner with the complete logic and interlock diagrams.

7.4 Design Requirements

- General

The turbine bypass control system shall be designed to provide maximum availability and reliability consistent with fail safe operation.

The system shall be of proven design, backed by field experience over multiple installations operating over a number of years.

- Operator Interface

The operator interface shall contain all necessary control and indication devices necessary for the operation of the function for which it is intended.

8. ELECTRICAL SPECIFICATION

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the turbine bypass system. This equipment shall include but not be limited to the following:

- Motors.
- All electrical control, monitoring and instrumentation devices and other

appurtenances required for this Specification M-302 “Turbine Bypass System” shall be provided by the Bidder.

8.2 Motors

For detail electrical requirements, see **Part 4, Section 3, E-210 “Low Voltage Motors”**.

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